

EXPERIENCE

- NetApp Inc.

Software Engineer Intern

Contributed to web platform development in TypeScript and PostgreSQL, implementing API endpoints, refactoring existing code, and reviewing pull requests

Wichita, KS
Jun 2026 – Present
- Software Engineer in Test

Built automated analysis pipeline processing storage telemetry across distributed RAID systems (FC, SAS, NVMe/RoCE), handling terabytes of performance data.

Designed Python automation framework reducing manual configuration tasks by 30% across Linux, Windows, and VMware infrastructure.

Implemented Ansible-based deployment orchestration for 300+ system configurations, streamlining infrastructure provisioning workflows.

Developed interactive visualization dashboard for system performance metrics using Python, analyzing 1M+ database entries for engineering insights.

Jul 2024 – Jun 2026

PROJECTS

- Leviathan

Searchless Transformer Chess Engine · PyTorch, FastAPI, Next.js, TypeScript, Tailwind CSS

Designed and trained 20M-parameter transformer with custom attention for real-time inference on edge hardware (RTX 4060, 8GB VRAM), optimizing architecture and training for resource-constrained deployment.

Reduced KV-cache memory footprint by 8× through Multi-Head Latent Attention implementation, enabling batch inference and 19ms per-move latency in real-time inference settings.

Designed custom tokenizer encoding board states and UCI notation into 4,247-token vocabulary for single-token move prediction.

Trained on 3M+ chess positions using gradient checkpointing and mixed-precision (FP16) to maximize data throughput under memory constraints.

HuggingFace Spaces
Jan 2025 – Dec 2025
- Multi-Head Latent Attention (MLA)

Research Implementation · PyTorch

Translated Multi-Head Latent Attention from DeepSeek-V2 research into modular PyTorch implementation, designing clean abstractions for KV compression and low-rank projection components.

Packaged implementation as production-ready PyPI library with type hints, documentation, and integration examples.

GitHub • PyPI
Jun 2025 – Aug 2025

EDUCATION

- Wichita State University

Accelerated Master of Science in Computer Science

Thesis: Generative Agent-Based Models for Insider Threat Detection

Relevant Coursework: NLP, Reinforcement Learning

Wichita, KS
Jan 2026 – May 2027 (Expected)
- Wichita State University

Bachelor of Science in Computer Science; GPA: 3.66/4.0, Magna Cum Laude

Dean’s List: Spring 2022 – Fall 2025

Relevant Coursework: Machine Learning, Artificial Intelligence, Fundamentals of AI Agents, Data Science

Wichita, KS
Jan 2022 – Dec 2025

SKILLS

- ML/DL: PyTorch, Transformers, Hugging Face, Model Training, Fine-tuning, Attention Mechanisms, Gradient Optimization

• Data & Computation: NumPy, Pandas, Scikit-learn, Matplotlib, Mixed-Precision Training

• Languages: Python, C++, C, SQL, JavaScript, TypeScript

• Tools & Systems: Git, Docker, Linux, Ansible, Jupyter, FastAPI